

Vedant Pol

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SKILLS

Languages: Python, SQL, Java, Bash

GenAI & LLMs: Retrieval-Augmented Generation, LangChain, LangGraph, LLM Evaluation, Prompt Engineering, LLM-as-Judge, Agentic Workflows, Guardrails, Prompt Injection Testing

Retrieval & Vector Search: Embeddings, Pinecone, Milvus, FAISS, Semantic Search, Hybrid Search, Reranking, Metadata Filtering, Chunking Strategies, Citation-Grounded Generation

ML, MLOps & Tools: PyTorch, Hugging Face Transformers, scikit-learn, MLflow, Docker, Kubernetes, AWS SageMaker, Kafka, Spark, Playwright, Pytest, Allure, GoCD

WORK EXPERIENCE

IDFC FIRST Bank, Mumbai

2024 - Present

Software Development Engineer in Test, Data Automation & GenAI QA

- Built data-driven automation frameworks for enterprise banking workflows across **payments, escrow, DigiX-CRO/CMS, and UPI**, validating transactions through expected-vs-actual checks, SQL/database validation, log analysis, and QA reporting.
- Automated transaction and data-quality validation by comparing source, database, API, and report outputs, strengthening regression coverage, defect traceability, and release confidence across payment workflows.
- Developed a **GenAI-powered QA Test Design Generator** using Python, FastAPI, Streamlit, SQLite, and OpenAI-compatible LLM APIs to convert FSDs, architecture docs, Jira issues, and Confluence pages into structured test cases.
- Generated traceability matrices, assumptions, coverage gaps, and downloadable **JSON/DOCX/XLSX** outputs while applying prompt engineering, LLM orchestration, document parsing, structured JSON generation, and large-document processing.
- Built practical DS/AI foundations through feature engineering, model evaluation, AWS, and MLOps learning, with interest in applying ML to fintech, fraud detection, document intelligence, payments, and intelligent automation.
- Python, SQL, FastAPI, Streamlit, SQLite, LLM APIs, Prompt Engineering, Document Parsing, Data Validation, Automation, AWS, MLOps, Banking Payments

Google Summer of Code, XWiki

Jun 2022 - Sep 2022

Software Engineering Intern, Open Source

Remote

- Built the **XWiki Snap package** and automated cloud-ready release workflows using CI/CD, Linux packaging, Docker, Kubernetes, and deployment automation.
- Contributed **24 PRs, 85 commits, and 7K+ lines of code** (Commits, Report); created user docs and developer docs for setup, builds, packaging, automation, and deployment.
- GitHub Actions, Linux Packaging, CI/CD, Docker, Kubernetes, Bash, Open Source, Developer Documentation

EDUCATION

Thakur College of Engineering & Technology

2020 - 2024

Bachelor of Engineering in Computer Science

CGPA: 9.08/10

Relevant Coursework: Data Structures and Algorithms, Databases, Operating Systems, Computer Networks, Machine Learning, Artificial Intelligence, Software Engineering, Cloud Computing

PROJECT WORK

- **Enterprise RAG Knowledge Assistant** (Live, GitHub): Built a GenAI assistant for policy and technical document search using semantic retrieval, metadata filtering, reranking, and citation-grounded answer generation. Implemented PDF parsing, text cleaning, chunking, embedding generation, vector indexing, and source-aware retrieval. Python, LangChain, Pinecone, FastAPI, Docker.
- **Multi-Agent Knowledge Management System** (Live, GitHub): Developed an agentic LLM workflow using LangGraph with Retriever, Planner, Critic, and Summarizer agents. Implemented guardrails for prompt injection, unsupported claims, citation validation, and hallucination detection using rule-based checks and LLM-as-Judge evaluation. LangGraph, AutoGen, FastAPI, Docker.
- **RetailPulse AI: Forecasting & Recommendation Platform** (Live, GitHub): Built and deployed an end-to-end ML platform for retail demand forecasting, anomaly detection, stockout risk scoring, product recommendations, model explainability, and AI-assisted business insights. Python, FastAPI, Streamlit, LightGBM, scikit-learn, Docker, MLflow-style tracking.

PATENT

- **ML-Based Dot Matrix Printer Fault Diagnosis** (German Patent): Developed an audio-based supervised ML system using Python, signal processing, feature extraction, classification models, and hardware sensor data to detect paper jams, missing print lines, carriage faults, and print head issues. Python, Machine Learning, Signal Processing, CNNs, Raspberry Pi.

AWARDS AND CERTIFICATES

- **AWS Certified Machine Learning – Specialty**: Credential
- **AWS Certified Cloud Practitioner**: Credential
- **Mathematics for Machine Learning, Imperial College London**: Certificate